

Study on Warpage of Flip Chip structure based on Finite Element Analysis and Machine Learning Algorithm

Qiwei Zhang#

Shenzhen Institute of Advanced
Electronic Materials, Shenzhen
Institute of Advanced Technology,
Chinese Academy of Science
Shenzhen 518055, China
qw.zhang2@siat.ac.cn

Cheng Zhong#

Shenzhen Institute of Advanced
Electronic Materials, Shenzhen
Institute of Advanced Technology,
Chinese Academy of Science
Shenzhen 518055, China
cheng.zhong1@siat.ac.cn

Sun Rong*

Shenzhen Institute of Advanced
Electronic Materials, Shenzhen
Institute of Advanced Technology,
Chinese Academy of Science
Shenzhen 518055, China
rong.sun@siat.ac.cn

Abstract—Warpage has always been a challenge in package, large value of warpage will lead to a series of problems such as delamination, crack et al. Finite element analysis (FEA) has been widely used in package, because of its comprehensive theory, diverse methods, convenient modification and low cost. Meanwhile, machine learning (ML) is good at constructing complex relationship between multiple features and target variables. Nowadays, the packaging structure is increasingly complex, and it is increasingly difficult to analyze and optimize directly according to the mechanical properties of materials, and ML is a potential solution. In this paper, we combine FEA with ML to study the warpage issue in a flip-chip structure. Three ML methods, polynomial regression, support vector machine (SVM), and artificial neural network (ANN) are used. Finally, the optimal results of ML are verified using FEA. A Flip Chip Ball Grid Array (FCBGA) structure was constructed in Abaqus and the CTE and Young's modulus of five materials were chosen as variables. The results show that, first of all, all of three ML models can excellently construct the mapping relationship between the ten variables of the ML model and the warpage value. The R-square index for all three models reaches above 0.999. In addition, traditional ML methods such as polynomial regression can achieve better results than SVM and ANN for warpage problems. Finally, it also indicates that compared to commonly used methods such as response surface methodology (RSM), the ML model can consider the impact of more variables on the results and optimize the warpage more effectively. This paper demonstrates the feasibility of using the ML algorithm for regression, prediction, and optimization of results in finite element analysis and applies it to warpage simulations.

Keywords—Warpage, Flip—chip, Finite element analysis (FEA), Machine learning (ML)

I. INTRODUCTION

In recent years, with the updating of electronic products, the integrated circuit manufacturing feature size continues to decrease, it is becoming increasingly difficult to reduce the feature size to meet the growing demand for electronic products. Therefore, in order to continue the effectiveness of Moore's law, the chip size is getting larger, and the package design and process are become more complex, which puts higher requirements for chip design, package process, material performance and reliability research [1]. Warpage has been one of the important causes of various failures, which is mainly due to the fact that when the temperature changes, the different coefficients of thermal expansion (CTE) of the materials of each component in the electronic package will lead to different degrees of deformation, and as the warpage

becomes larger, the stability of the whole package structure will become worse, which will cause the occurrence of other failure modes, such as delamination, crack et al [2]. Even with the increasing trend of chip and package size, the warpage problem will be more serious, and due to the increasing complexity of the package structure design, it becomes very difficult to improve the warpage performance by analyzing the mechanical properties of different materials directly.

For the above problems, machine learning methods can be a potential solution given their strong ability to construct complex relationships between multiple features and target variables. And in existing research, there has been many works that use machine learning to optimize packaging design [3-7]. In this paper, we study the warpage under Flip Chip Ball Grid Array (FCBGA) package, construct FCBGA package model with experimental data through Abaqus finite element analysis (FEA) software, use three machine learning algorithms to directly construct the mapping relationship between multivariate and warpage, and obtain the optimal parameter combinations to minimize warpage according to the obtained machine learning model, to achieve the purpose of optimizing package warpage while avoiding complex mechanical analysis. Firstly, we constructed a FCBGA packaging model through Abaqus software, selecting the elastic modulus and CTE of five materials in the model as variables. Then, an orthogonal experimental design was conducted based on a reasonable range of material property. According to the results of the orthogonal experimental design, 64 combinations of features are generated as the training dataset for the machine learning model, which ensures that the feature space covers the entire value range. Next, ten combinations were designed as the testing dataset for the machine learning model, and apply the FCBGA finite element model (FEM) to all combinations from training and test datasets to obtain the substrate side warpage values as the predicted target variables. Secondly, machine learning models were trained using three different algorithms, and R-squared decision coefficients, mean squared error (MSE) and mean absolute error (MAE) are used as model evaluation indexes. Finally, the Cartesian product combinations of five equally spaced values within the range of material parameter values yielded a total of 9,765,625 parameter combinations, which then were predicted using trained machine learning model. Then the best parameter combinations with the minimum predicted warpage value are simulated by FEM to verify whether the machine learning algorithm can optimize the warpage value of the package model.

II. MODELLING

Fig. 1 shows the flowchart of for optimizing warpage using machine learning methods. Each step in the flowchart will be discussed in detail in the following sections.

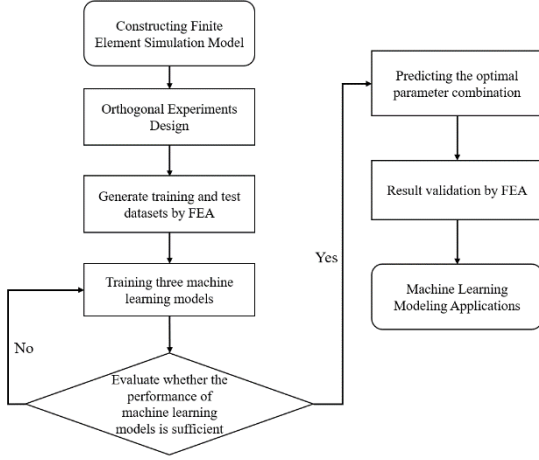


Fig. 1. Machine learning optimization warpage method flowchart

A. Finite Element Modelling

The standard Lid cover FCBGA structure is shown in Fig. 2, including Lid, Thermal interface material (Tim), Silicon die, Underfill, Solder, Adhesive, Substrate seven parts, for simplicity we regard Underfill and Solder as the same material Underfill. And since the mechanical properties of the silicon die are basically unchanged, we select the CTE and elastic modulus of the rest of five materials as variables.

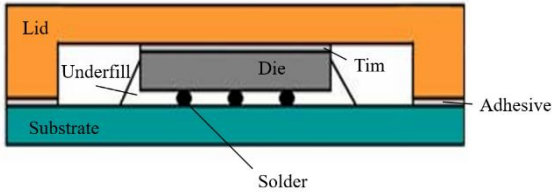


Fig. 2. FCBGA with lid package structure

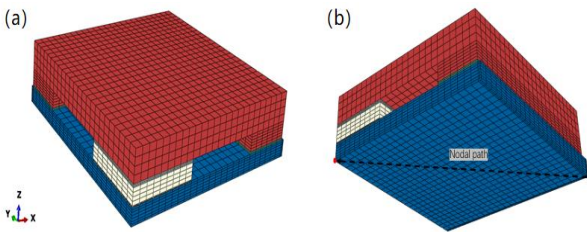


Fig. 3. (a) Quarter symmetric FEA model of flip-chip package, (b) the nodal path for extracting warpage at the bottom of substrate.

A 1/4 symmetric FCBGA with lid package finite element model was constructed using Abaqus software, the size of substrate is 25×25 mm², the material was set to be an isotropic material, the temperature load setting from 25°C to 150°C and all elements type using C3D8R. The FEM is shown in Fig. 3(a). The model warpage values are extracted by the z direction displacement values on the diagonal path of the baseplate bottom surface as shown in Fig. 3(b).

B. Data Preparation

1) Data Preprocessing

For machine learning models, we need to pre-process the input data to avoid the negative impact on the model due to the different scale of the original data. For example, the modulus of elasticity of silicon is 131000 Mpa and the CTE is 2.8 ppm/°C. Numerically, the difference is very large, if the data is not preprocessed, the modulus change will have a greater impact on the model. Therefore, we need to normalize the data, which is to map the original data to the range of [0,1], the specific formula was shown in (1):

$$y = \frac{X - \min(X)}{\max(X) - \min(X)} \quad (1)$$

Where y is the new generated data, X is the original data, and min(X) and max(X) represent the maximum and minimum values of the original features, respectively. It is worth to note that in this paper, the data processing and model are implemented in python language, and the algorithm calls the open source function library scikit—learn [8].

2) Orthogonal Experimental Design

In this paper, ten mechanical property parameter variables were selected according to the FCBGA package model as respectively Lid_CTE, Lid_E, Tim_CTE, Tim_E, Underfill_CTE, Underfill_E, Adhesive_CTE, Adhesive_E, Substrate_CTE, Substrate_E. And a proper parameter range is selected for each material, three or four values are taken as inputs of the parameters in the range with equal spacing respectively, the parameter values of each material after the normalization process are shown in Table 1.

TABLE I. NORMALIZATION RANGE TABLE

Features	Normalized value point
Lid_CTE	[0, 0.333, 0.667, 1]
Lid_E	[0, 0.5, 1]
Adhesive_CTE	[0, 0.5, 1]
Adhesive_E	[0, 0.5, 1]
Tim_CTE	[0, 0.5, 1]
Tim_E	[0, 0.5, 1]
Underfill_CTE	[0, 0.5, 1]
Underfill_E	[0, 0.5, 1]
Sub_CTE	[0, 0.333, 0.667, 1]
Sub_E	[0, 0.333, 0.667, 1]

After determining the parameter values, orthogonal experimental design is used to ensure that the combinations of feature point values can cover all the feature space, and a total of 64 different feature combinations are generated. Then Abaqus software is used to simulate these 64 feature combinations to obtain the corresponding substrate warpage values for each parameter combination, and these 64 simulations are used as the training datasets of the machine learning model, and the first ten sets of normalized input data and warpage results of the cases in all the training datasets are shown in Table II.

TABLE II. FIRST TEN TRAINING DATASETS

Case number	Input features	Warpage (um)
1	[1, 0, 0, 0, 0, 0, 0, 0, 1, 1]	32.71
2	[0.667, 0.5, 0.5, 0, 0, 0, 0, 0.5, 1, 0.333]	31.06

3	[0.667, 0, 0, 0.5, 0, 1, 0, 0.5, 0, 0.667]	14.28
4	[0, 0.5, 0, 1, 0, 0, 0.5, 1, 0.333, 0]	19.52
5	[0.667, 0.5, 0, 0, 0, 1, 0.5, 0, 0.333, 0.667]	19.47
6	[0.333, 0, 0, 1, 0, 0.5, 0, 1, 1, 0.333]	30.35
7	[0, 0, 0, 1, 0.5, 0.5, 1, 0, 0.333, 1]	20.55
8	[1, 0, 0, 0.5, 1, 1, 0.5, 0, 0.667, 0]	22.71
9	[1, 0.5, 0, 0.5, 0, 0.5, 0, 0, 0.667, 0.667]	25.25
10	[0.333, 1, 0, 1, 0.5, 1, 0.5, 0.5, 1, 1]	31.25

In addition to the training dataset, we also need to construct a test dataset to evaluate the predictive ability of the trained machine learning model. In order to avoid the data used in the training dataset, we took 12 points equally spaced within the parameter range and removed the two endpoints for linear combination, and finally obtained ten test data, which were simulated by FEM to obtain the corresponding warping values, and the normalized test dataset are shown in Table III.

TABLE III. TEST DATASETS

Case number	Input features	Warpage (um)
1	[0.1, 0.09, 0.09, 0.1, 0.09, 0.09, 0.09, 0.09, 0.09]	15.50
2	[0.17, 0.18, 0.18, 0.2, 0.18, 0.18, 0.18, 0.18, 0.18]	17.09
3	[0.27, 0.27, 0.28, 0.25, 0.28, 0.28, 0.28, 0.27, 0.28, 0.27]	18.70
4	[0.37, 0.36, 0.37, 0.35, 0.37, 0.36, 0.37, 0.36, 0.37, 0.36]	20.32
5	[0.47, 0.45, 0.46, 0.45, 0.46, 0.46, 0.46, 0.45, 0.46, 0.45]	21.94
6	[0.53, 0.55, 0.55, 0.55, 0.55, 0.54, 0.55, 0.55, 0.54, 0.55]	23.43
7	[0.63, 0.64, 0.64, 0.65, 0.63, 0.64, 0.64, 0.64, 0.63, 0.64]	25.06
8	[0.73, 0.73, 0.73, 0.75, 0.73, 0.73, 0.73, 0.73, 0.72, 0.73]	26.69
9	[0.83, 0.82, 0.82, 0.8, 0.82, 0.82, 0.82, 0.82, 0.82, 0.82]	28.33
10	[0.9, 0.91, 0.91, 0.9, 0.91, 0.91, 0.91, 0.91, 0.91, 0.91]	29.96

3) Evaluation indicators

We need to make a reasonable evaluation of the prediction performance of the trained machine learning model. In this paper we use mean square error (MSE), mean absolute error (MAE), and R-Square (R2) decision coefficients as evaluation indexes to assess the accuracy of the regression model, and they are all commonly used to evaluate the effectiveness of regression models, the specific formulas are shown in (2), (3), and (4).

$$MSE = \frac{1}{m} \sum_{i=1}^m (y_i - \hat{y}_i)^2 \quad (2)$$

$$MAE = \frac{1}{m} \sum_{i=1}^m |y_i - \hat{y}_i| \quad (3)$$

$$R2 = 1 - \frac{\sum_i (y_i - \bar{y}_i)^2}{\sum_i (\bar{y}_i - \bar{y}_i)^2} \quad (4)$$

Where y_i is the true value, $\hat{y}_i$ is the predicted value, $\bar{y}_i$ is the average value of the true value, the R2 index compared to MAE and MSE, more effectively reduce the impact of the outliers in the evaluation of the prediction effect, mainly reflecting the proportion of the prediction error value

compared to the standard value of the error, which is used as the main index of the model evaluation.

C. Polynomial Regression

Polynomial regression is one of the most widely used regression algorithms, and in fact the commonly used response surface method (RSM) also is a polynomial regression model, but related studies are generally limited to the use of binary-quadratic regression in order to obtain a visualized response surface [3, 4], which can be written as a regression equation as shown in (5), containing an intercept and five coefficients.

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1 x_2 + \beta_4 x_1^2 + \beta_5 x_2^2 \quad (5)$$

In contrast, we directly use all ten parameters as features for constructing the quadratic regression equation, which can consider the effect of all variables on warpage and has better regression prediction ability. The constructed regression equation contains an intercept and 66 coefficients.

The principle of the regression equation is based on the least squares method to solving (6) by matrix transformation. Where y is the true value matrix, X is the matrix consisting of the training set, and ω , the parameter matrix we require to solve, has a shape of (67, 1). Finally the ω matrix that minimizes (6) can be found.

$$\min_{\omega} \|y - X\omega\|_2^2 \quad (6)$$

D. Support Vector Machine

Support Vector Machine (SVM) is a commonly used algorithm in the field of machine learning, which shows better performance in the processing of small samples and nonlinear problems. The core idea is to map the linearly indistinguishable vector data in the low-dimensional space to the high-dimensional space, and then find a hyperplane in the high-dimensional space can be a good distinction between these data, and the vectors used to construct the hyperplane are called support vectors. A two-dimensional space hyperplane schematic was shown in Fig. 4(a). The SVM idea can also be used for regression analysis, unlike the SVM which aims to classify the data, the SVR aims to find a hyperplane that can best contain all the data within a certain range which is illustrated in Fig. 4(b). Therefore, the actual problem is to be solved as shown in (7):

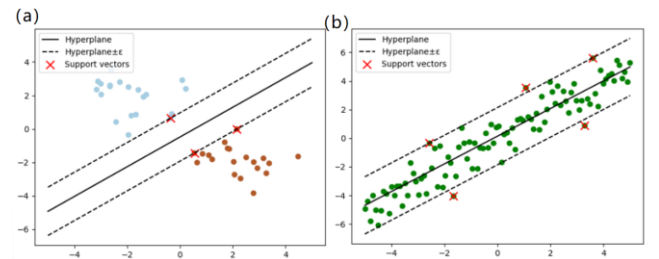


Fig. 4. Schematic diagram of (a) support vector machine and (b) support vector regression

$$\min_{\omega, \xi, \xi^*} \frac{1}{2} \omega^T \omega + C \sum_{i=1}^n (\xi_i + \xi_i^*) \quad (7)$$

$$s.t. \quad y_i - \omega^T \phi(x_i) - b \leq \varepsilon + \xi_i,$$

$$\omega^T \phi(x_i) + b - y_i \leq \varepsilon + \xi_i^*,$$

$$\xi_i, \xi_i^* \geq 0, i=1, \dots, n$$

Where x_i means training dataset parameter vectors, y_i means target parameter vectors. C means the penalty coefficient, ε means the range of training dataset vectors contained in the hyperplane, and ξ means the value of vectors that are out of the range of ε in the hyperplane, because there are two cases of exceeding the hyperplane region with positive and negative values, so there are two constraint equations. Briefly, the equations penalize the vectors whose distance from the hyperplane exceeds the range of ε , thus obtaining the ω matrix that minimizes the error.

E. Artificial Neural Network

Artificial neural network (ANN), also be called multilayer perceptual machines is has been a hot research field in recent years, and its variants convolutional neural networks, recurrent neural networks in computer vision, natural language processing and other fields have emerged very fascinating results. Its core idea has been to construct the mapping relationship between input and output by training the network structure, and this method has shown better results in solving nonlinear problems. Therefore, in this paper, we constructed a fully connected artificial neural network structure with ten material variables as inputs and warpage values as outputs, and evaluated the prediction performance of the ANN regression model.

The main steps of ANN are as follows, firstly, the neural network architecture is constructed, including the input, hidden and output layers, then the relevant key hyperparameters are selected, such as the activation function, the learning rate et al, then calculated the loss function, and finally, the parameters of each neuron in the model are updated according to the value of the loss function by forward propagation, and the process is repeated until the model converges. A neural network schematic is shown in Fig. 5, and the computation of each neuron in a neural network is shown in (8), where f is the activation function, ω_i and x_i are the inputs of the previous layer and their respective weights, and b is the neuron intercept term.

$$y = f(\sum_{i=1}^n \omega_i x_i + b) \quad (8)$$

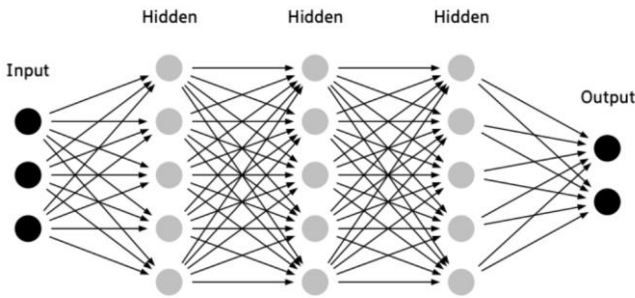


Fig. 5. Schematic diagram of neural network

III. RESULT AND DISCUSSION

In this section, we trained regression models for three machine learning methods, polynomial regression, support vector machine and artificial neural network, using the

training dataset and test dataset generated in the previous section. Then the model prediction effectiveness was evaluated using R-square coefficients, MSE and MAE and the strengths and weaknesses of the models were compared. Finally, all combinations in the parameter range were traversed by the Cartesian product to find the combination of mechanical property parameters that minimized the predicted warpage value, and the results were validated using a FEM, demonstrating the effectiveness of optimizing the package warpage using machine learning algorithms.

A. Polynomial Regression Model

We constructed our polynomial regression model using a quadratic multiple regression analysis with a training dataset which contained 64 data points. A regression equation containing an intercept and 66 coefficients is obtained. The results of the test using the test dataset are shown in Fig. 6, where the black points true data and the blue lines indicate the model prediction curve. The predictive effectiveness of the regression model was also evaluated, and the results for each metric are shown in Table IV. The R-2 score of 0.99996 indicates that the model can almost perfectly predict the warpage values in the test dataset based on the mechanical parameters of the material, while the whole process takes only 0.006 seconds.

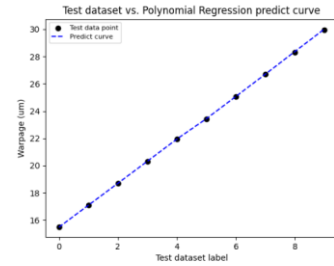


Fig. 6. The Polynomial Regression predicted data compared to test data of warpage

TABLE IV. RESULT OF POLYNOMIAL REGRESSION MODEL

Evaluation indexes	R2	MSE	MAE	Time (s)
Value	0.99996	0.0009	0.021	0.006

In previous studies, the response surface method often just uses 2-3 key parameters for quadratic regression, while in this paper all the variables are used for model training, in order to explore what kind of effect reducing the features in the model training process will have, we only use all the material CTE and all the material modulus, respectively, i.e., the original ten parameters are reduced to five parameters for model training, and the parameters of the regression equation are also reduced from 67 to 22, and the model evaluation effect is as shown in Fig. 7(a) and (b), and Tables V.

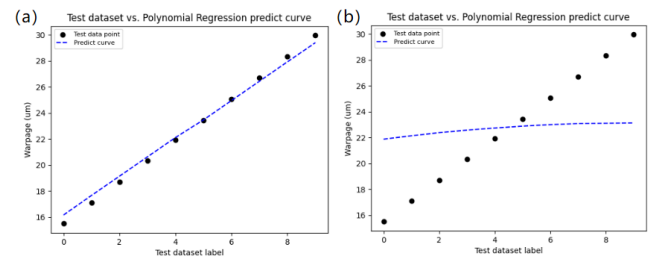


Fig. 7. The polynomial regression model predicted warpage compared true warpage with trained with (a) CTE and (b) modulus

TABLE V. RESULT OF POLYNOMIAL REGRESSION MODEL

Evaluation indexes	R2	MSE	MAE	Time (s)
All CTE model score	0.992	0.16	0.35	0.007
All E model score	0.166	17.68	3.64	0.006

It can be seen that, consistent with our original understanding, the change of warpage value is mainly caused by CTE mismatch between materials. The model constructed by modulus is far less effective than the model constructed by CTE. However, the two models with reduced variables are not as effective as the model using all ten variables. This is because the reduction in training features also results in the loss of some variable information related to the final warpage value. Therefore, we believe that when constructing the mapping relationship between features and target variables, the more information, the better result.

B. Support Vector Regression (SVR)

We use a similar approach to construct the support vector regression model, the hyperparameters of the regression model are determined using a grid search method, the kernel function is chosen the Radial basis kernel function (RBF), the kernel parameter is set to 0.01, the penalization coefficient C and the hyperplane unpenalized range ϵ are 1000 and 0.1, and the hyperparameters have been explained in Section II.D. The model prediction result and evaluation are shown in Fig. 8 and Table VI. The same reduction of features only uses CTE and modulus to evaluate the training effectiveness, result is the same and has shown in Fig. 9 and Table VII.

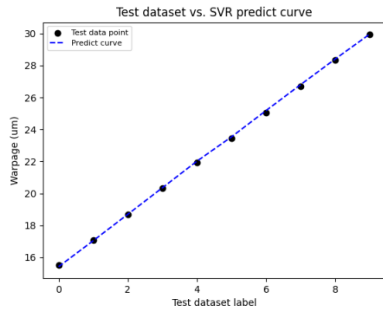


Fig. 8. The SVR predicted data compared to test data of warpage

TABLE VI. RESULT OF SVR MODEL

Evaluation indexes	R2	MSE	MAE	Time (s)
Value	0.9996	0.007	0.07	0.008

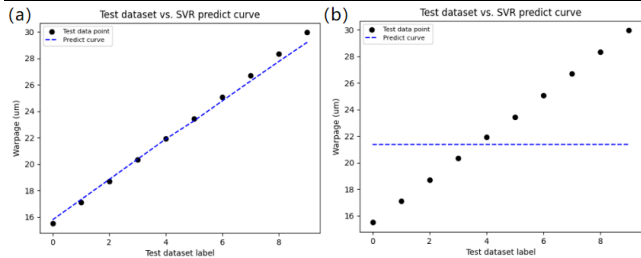


Fig. 9. The SVR predicted warpage compared true warpage with trained with (a) CTE and (b) modulus

TABLE VII. RESULT OF SVR MODEL

Evaluation indexes	R2	MSE	MAE	Time (s)
All CTE model score	0.994	0.127	0.286	0.005
All E model score	-0.08	22.95	4.1	0.006

C. Artificial Neural Network

We also chose the grid search method to select the hyperparameters for the ANN regression model. Finally, we constructed an artificial neural network containing three hidden layers, plus the input and output layers in a total of five layers of fully connected structure. Each hidden layer contains 50 neurons, the Rectified Linear Unit (relu) is chosen as the activation function, and the quasi-Newton methods “lbfgs” are chosen as the solver to optimize the loss function. This method shows faster and better performance for small sample problems and the model converges after 1099 iterations. The results are shown in Fig. 10 and Table VIII.

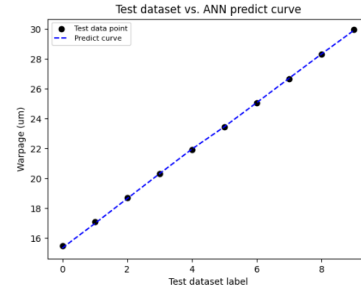


Fig. 10. The ANN predicted data compared to test data of warpage

TABLE VIII. RESULT OF ANN MODEL

Evaluation indexes	R2	MSE	MAE	Time (s)
Value	0.9998	0.005	0.05	0.514

Also only using the material CTE or modulus to train this model for prediction the results are shown in Fig. 11 with Tables IX, with no change in the conclusions.

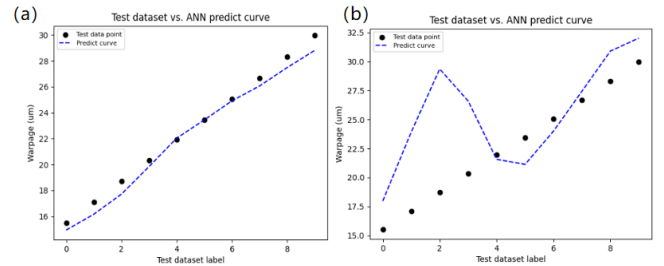


Fig. 11. The ANN predicted warpage compared true warpage with trained with (a) CTE and (b) modulus

TABLE IX. RESULT OF ANN MODEL

Evaluation indexes	R2	MSE	MAE	Time (s)
All CTE model score	0.977	0.478	0.58	1.465
All E model score	-0.06	22.4	3.54	3.45

Overall, all three machine learning models show excellent results in their ability to predict FEM warpage using material parameters as inputs. From the evaluation index, polynomial regression achieved higher R-2 score than both the SVM and ANN method, which was not anticipated. We believe that this is mainly due to three reasons, firstly, the FCBGA package FEM we have constructed for warpage value variation does not have very strong nonlinear characteristics, so the support vector machine and artificial neural network methods with stronger nonlinear capabilities are instead less effective than the simpler polynomial regression. Second, our training dataset is small compared to datasets in the computer field, and the contrast presented for complex algorithms versus simple algorithms is not obvious. Finally, it should be noted

that the R2 scores for all three models reach above 0.999, so in fact all three models perform very well and there is no big difference between them. The polynomial regression does show better performance for warpage prediction based on the FEM we constructed. However, we believe that more sophisticated methods such as SVM and ANN can show better performance if there is a strong nonlinear relationship between the target variable and the input features, similar performance was shown in [5].

D. Optimize Design Parameters

After constructing a regression model through training data, we hope to obtain the optimal material parameters to minimize the packaging warpage value. Therefore, we take five points at equal intervals for each material in the previous range of material values, and then do Cartesian product for a total of 9765625 parameter combinations. Then use machine learning models to predict them to get the optimal parameter combination. Finally, use a FEM to verify whether this parameter combination can achieve very good warpage results. The optimal parameter combinations obtained for the three machine learning models with the corresponding results and time costs are shown in Table X.

TABLE X. OPTIMIZATION RESULTS

Model	Polynomial Regression	SVR	ANN
Lid_CTE (ppm/°C)	17	17	17
Lid_E (Mpa)	130000	110000	130000
Adhesive_CTE (ppm/°C)	110	130	130
Adhesive_E (Mpa)	6.2	8.2	8.2
Tim_CTE (ppm/°C)	120	120	120
Tim_E (Mpa)	50	50	50
Underfill_CTE (ppm/°C)	90	90	70
Underfill_E (Mpa)	8000	6000	6000
Sub_CTE (ppm/°C)	13	13	13
Sub_E (Mpa)	14000	14000	14000
Predicted warpage value (um)	10.59	12.36	12.66
Simulated Warpage Value (um)	13.49	12.95	12.72
Cartesian product combinations predict time (s)	1477.2	1005.2	892.3

The optimal material parameter combinations obtained from the three models were validated by FEA with optimal warpage values of 13.5 um, 12.9 um and 12.7 um, respectively. All three warpage results are very good, with each value ranking in the top two of all previous data, and the best parameter combination obtained by the ANN model corresponds to a warpage value of 12.7 um even smaller than the minimum value of 12.8 um in the training dataset. Therefore, material parameter combinations can be designed through ML models to optimize packaging warpage values while avoiding complex mechanical analysis of the structure. And using three models to predict 9765625 parameter combinations takes less than half an hour, indicating that ML models have greatly improved the efficiency of prediction compared to finite element simulation, as the model can accurately reflect the target variable.

We also implemented a gradient descent algorithm to minimize the ML model as well as to get the corresponding

input parameter combinations, and the optimal parameter combinations obtained are almost the same as the results obtained by the Cartesian product method. Among all the parameters, only the Underfill_E value of the polynomial regression model is changed from 8000 Mpa to 7531 Mpa, while the corresponding predicted warpage value is 10.58 um and the simulated warpage value is 13.42 um. Not only better results were obtained, the time needed to optimize the three models was also greatly reduced, the three models time costs were 0.035s, 0.006s and 0.004s, respectively.

IV. CONCLUSIONS

We constructed a machine learning dataset based on the FEM of FCBGA package and trained three machine learning models to optimize the design of the material parameters, the results were verified by FEA. The results show that the machine learning model can construct the relationship between the mechanical properties of the material and the warpage very well, and when more features are used to train the machine learning, the better the model performance. For the warpage problem of the FCBGA package model, the polynomial regression model shows better performance than SVM and ANN. Finally, we optimized the material parameter combinations based on the trained models, and all the models achieved relatively good results, in which the material parameters obtained from the ANN even got better results than all the previous data. Therefore, the machine learning algorithm can avoid the complex analysis of the mechanical properties of the packaging structure, and directly get the optimized material parameter combinations, and the machine learning model also has obvious advantages in its computational efficiency compared to the finite element simulation.

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